

Is Learnable Attention Normalization Beneficial? An Empirical Investigation with Mixed Findings

Muhammad Maroof Farooq
Independent Researcher, Karachi, Pakistan
maroof96965@gmail.com

April 2026 *Workshop submission — under review*

Abstract

The scaled dot-product attention mechanism at the core of every Transformer architecture applies a *fixed* softmax normalization uniformly across all heads, layers, and inputs. A natural question is whether replacing this fixed function with a learnable, input-conditioned alternative can improve model expressiveness or performance. We study this question empirically by introducing **Adaptive Activation Attention** (A^3), a framework in which the attention normalization is parameterized by a lightweight gated transformation that conditions on the current input. The construction satisfies non-negativity and normalization by design, and is initialized to reproduce standard softmax behavior.

We conduct small-scale experiments on text classification (SST-2, MNLI), named entity recognition (CoNLL-2003), and a synthetic copying task, comparing A^3 to softmax, learned-temperature softmax, sparsemax, and α -entmax baselines, each run over five random seeds. **Our findings do not support the hypothesis that learnable normalization provides reliable benefit over a well-tuned softmax baseline.** Observed improvements are small (at most +0.9 F1 on CoNLL-2003), statistically inconclusive on most tasks when assessed with bootstrap confidence intervals, and sensitive to initialization and learning rate schedules. On two of five seeds for MNLI, A^3 underperforms standard softmax. A learned-temperature softmax, which adds only one scalar parameter per head, matches A^3 on every classification task.

We document three concrete failure modes—gate collapse, gradient spiking on harder tasks, and uniform gate convergence that collapses to learned temperature—and analyze the fraction of heads that actually exploit the input-conditioning capacity. Roughly half the heads converge to near-constant gate values, suggesting that the model frequently does not need or exploit dynamic normalization. We present these results as an honest negative-leaning empirical study and discuss what conditions would need to hold for learnable normalization to be worth pursuing at scale.

1 Introduction

The Transformer [Vaswani et al., 2017] has become the dominant architecture across NLP [Devlin et al., 2019, Brown et al., 2020], vision [Dosovitskiy et al., 2021], and multimodal learning. Its central operation is scaled dot-product attention:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}, \quad (1)$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d_k}$ and $\mathbf{V} \in \mathbb{R}^{n \times d_v}$. Every head in every layer applies the same softmax, regardless of input content, layer depth, or the functional role of the head.

This uniformity raises a reasonable question: is the fixed softmax optimal, or could a learned, input-dependent normalization improve the expressiveness of attention? Prior work has explored fixed alternatives—sparsemax [Martins & Astudillo, 2016] and α -entmax [Peters et al., 2019]—but these remain static functions that do not adapt at inference time. The most expressive prior approach, adaptively sparse Transformers [Correia et al., 2019], learns a scalar α per head but fixes it across inputs.

This work. We propose A^3 (**A**daptive **A**ctivation **A**ttention), which conditions the normalization on the current input via a lightweight gated score transformation. The design is mathematically principled: validity is guaranteed by construction, and the method recovers softmax at initialization. These properties make it a clean experimental testbed for the question of whether dynamic normalization is useful.

What we find. Despite the appealing motivation, our experiments reveal that A^3 *does not reliably outperform softmax*. Improvements are small, statistically inconclusive on most tasks, inconsistent across seeds, and easily matched by a per-head learned temperature. We document failure modes and analyze why the model often fails to exploit the input-conditioning capacity we provide. We present these findings not as a negative result in the pejorative sense, but as an honest contribution to understanding what softmax attention does and does not lack.

Why this matters. The field has a well-documented publication bias toward successful modifications to Transformer components. Careful, multi-seed empirical studies that find softmax to be surprisingly robust—and that document precisely *why* proposed alternatives fail—are underrepresented in the literature. We hope this study contributes to a more accurate empirical picture.

Contributions.

1. A mathematically valid framework for learnable attention normalization (A^3) with a proved softmax-recovery initialization (§4).
2. A multi-seed experimental evaluation with explicit bootstrap confidence intervals, including tasks and seeds where the method does not provide benefit (§6–7).
3. A documented taxonomy of failure modes with mechanistic explanations (§8).
4. Analysis of how much input-conditioning capacity is actually exploited at convergence (§9).

2 Related Work

2.1 Fixed Alternative Normalizations

Sparsemax [Martins & Astudillo, 2016] is the Euclidean projection onto the probability simplex:

$$\text{sparsemax}(\mathbf{z}) = \arg \min_{\mathbf{p} \in \Delta^{n-1}} \|\mathbf{p} - \mathbf{z}\|^2, \quad (2)$$

which assigns exact zero probability to low-scoring positions. The original paper reports improvements on certain tasks; subsequent replications find the gains to be task-dependent and not uniform.

α -entmax [Peters et al., 2019] defines a family parameterized by $\alpha \geq 1$:

$$\alpha\text{-entmax}_\alpha(\mathbf{z}) = \arg \max_{\mathbf{p} \in \Delta^{n-1}} [\mathbf{p}^\top \mathbf{z} + H_\alpha(\mathbf{p})], \quad (3)$$

where H_α is the Tsallis α -entropy, recovering softmax at $\alpha = 1$ and sparsemax at $\alpha = 2$. Correia et al. [2019] extend this by learning α per head. The key distinction is that their α is a fixed scalar at inference time, whereas our gate vector varies per input instance.

Temperature scaling is the simplest learnable extension of softmax, multiplying logits by a learned scalar $\tau_h > 0$ per head. This baseline is often omitted from comparisons in the literature, yet our results show it to be highly competitive. We include it as a primary baseline throughout.

2.2 Linear and Efficient Attention

Performer [Choromanski et al., 2021], cosFormer [Qin et al., 2022], and related methods replace softmax to reduce complexity from $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$. These target efficiency rather than expressiveness and are orthogonal to our investigation, which maintains full quadratic attention.

2.3 Structural Attention Modifications

Wu et al. [2019] and Child et al. [2019] modify which positions attend to which without changing the normalization. Our work is orthogonal: we fix the standard $\mathbf{QK}^\top$ score computation and modify only the downstream normalization step.

2.4 Why Softmax Persists

A complete treatment of this topic must acknowledge the reasons softmax has not been displaced despite a decade of proposed alternatives:

1. **Scale validation.** Softmax has been validated in models with hundreds of billions of parameters trained on trillions of tokens. Proposed alternatives have generally been tested only at small scale.
2. **Gradient well-posedness.** The softmax Jacobian has a known, stable structure. Sparse normalizations introduce zero-gradient regions that can destabilize training [Martins & Astudillo, 2016].
3. **Implicit sharpness control.** The $\mathbf{W}^Q$ and $\mathbf{W}^K$ projections can learn to amplify or suppress score magnitudes, providing effective sharpness control without any modification to the normalization function itself.

These observations set a high bar for what a proposed replacement must overcome. Our results suggest A³, in its current form, does not clear this bar.

3 Background

3.1 Standard Multi-Head Attention

Given $\mathbf{X} \in \mathbb{R}^{n \times d}$:

$$\mathbf{Q}_h = \mathbf{X}\mathbf{W}_h^Q, \quad \mathbf{K}_h = \mathbf{X}\mathbf{W}_h^K, \quad \mathbf{V}_h = \mathbf{X}\mathbf{W}_h^V, \quad (4)$$

$$\text{head}_h = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}}\right) \mathbf{V}_h, \quad (5)$$

$$\text{MHA}(\mathbf{X}) = \text{concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O. \quad (6)$$

The softmax in the second line is identical for all $h \in \{1, \dots, H\}$ and all layers ℓ .

3.2 Conditions for a Valid Attention Activation

Definition 1 (Valid Attention Activation). *A function $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ is a valid attention activation if, for every row i of its input $\mathbf{S} \in \mathbb{R}^{n \times n}$:*

1. $f(\mathbf{S})_{i,j} \geq 0$ for all j (non-negativity)
2. $\sum_j f(\mathbf{S})_{i,j} = 1$ (normalization)
3. f is differentiable almost everywhere (trainability)

A learnable f_θ must satisfy all three conditions for all values of θ encountered during training, not only at convergence.

4 The A³ Framework

4.1 Motivation and Design Principles

We construct a parametric attention activation satisfying four properties: (a) satisfies Definition 1 unconditionally for all θ ; (b) conditions on the current input at inference time; (c) requires $\mathcal{O}(d_k^2)$ additional parameters per head, not $\mathcal{O}(n^2)$; and (d) reduces to standard softmax at a natural initialization. Properties (a) and (d) together ensure that training begins from a well-understood baseline and cannot produce invalid attention maps at any point during optimization.

4.2 Gated Score Transformation

Let $\mathbf{S} = \mathbf{Q}_h \mathbf{K}_h^\top / \sqrt{d_k} \in \mathbb{R}^{n \times n}$ be the pre-normalization score matrix for head h in layer ℓ . The A³ activation is:

$$f_\theta(\mathbf{S}, \mathbf{c}) = \text{softmax}(\mathbf{S} \odot \mathbf{G} + \mathbf{B}), \quad (7)$$

where $\odot$ is element-wise multiplication. The context vector $\mathbf{c}$, gate $\mathbf{G}$, and bias $\mathbf{B}$ are:

$$\mathbf{c} = [\bar{\mathbf{Q}}_h; \bar{\mathbf{K}}_h; \mathbf{e}_{\ell,h}] \in \mathbb{R}^{2d_k + d_e}, \quad (8)$$

$$\mathbf{g} = \sigma(\mathbf{W}_g \mathbf{c} + \mathbf{b}_g) \in (0, 1)^{d_k}, \quad (9)$$

$$\mathbf{G} = \text{expand}(\mathbf{g}) \in \mathbb{R}^{n \times n}, \quad (10)$$

$$\mathbf{B} = \mathbf{U} \mathbf{c}^\top \in \mathbb{R}^{n \times n}, \quad (11)$$

where $\bar{\mathbf{Q}}_h = \frac{1}{n} \sum_i \mathbf{Q}_{h,i}$, and $\bar{\mathbf{K}}_h = \frac{1}{n} \sum_i \mathbf{K}_{h,i}$, are mean-pooled query and key vectors; $\mathbf{e}_{\ell,h} \in \mathbb{R}^{d_e}$ is a learned embedding identifying the layer and head pair; σ is the sigmoid function; $\mathbf{W}_g \in \mathbb{R}^{d_k \times (2d_k + d_e)}$; $\mathbf{U} \in \mathbb{R}^{n \times (2d_k + d_e)}$.

Semantics. A gate value $\mathbf{g}_j \approx 1$ preserves the j -th component of the score; $\mathbf{g}_j \approx 0$ suppresses it toward zero. Since the downstream softmax normalizes, suppressing all scores produces near-uniform attention while amplifying score contrasts produces sharper attention. The gate thus provides input-conditioned sharpness control. The bias $\mathbf{B}$ adds an input-dependent additive offset, allowing structured positional emphasis not captured by the multiplicative gate.

4.3 Validity and Initialization

Proposition 1 (Unconditional Validity). *For all θ and all inputs $(\mathbf{S}, \mathbf{c})$, the function f_θ in Eq. (7) is a valid attention activation.*

Proof. Fix any row i . Let $\tilde{\mathbf{S}}_{i,\cdot} = \mathbf{S}_{i,\cdot} \odot \mathbf{G}_{i,\cdot} + \mathbf{B}_{i,\cdot} \in \mathbb{R}^n$. Then $f_\theta(\mathbf{S}, \mathbf{c})_{i,\cdot} = \text{softmax}(\tilde{\mathbf{S}}_{i,\cdot})$, which maps any real vector to $\Delta_+^{n-1} \subset \mathbb{R}^n$. Non-negativity and normalization follow from softmax. Differentiability follows from the composition of σ (smooth), affine maps (smooth), and softmax (smooth). $\square$ $\square$

Proposition 2 (Softmax Recovery). *Initialize $\mathbf{W}_g = \mathbf{0}$, $\mathbf{b}_g = c \cdot \mathbf{1}$ for $c \gg 0$, $\mathbf{U} = \mathbf{0}$, $\mathbf{e}_{\ell,h} = \mathbf{0}$. Then $f_\theta(\mathbf{S}, \mathbf{c}) \approx \text{softmax}(\mathbf{S})$.*

Proof. With these values: $\mathbf{g} = \sigma(c \cdot \mathbf{1}) \approx \mathbf{1}$, so $\mathbf{G} \approx \mathbf{1}_{n \times n}$ and $\mathbf{B} = \mathbf{0}$. Thus $\mathbf{S} \odot \mathbf{G} + \mathbf{B} \approx \mathbf{S}$, giving $f_\theta(\mathbf{S}, \mathbf{c}) \approx \text{softmax}(\mathbf{S})$. $\square$ $\square$

Remark 1. *In all experiments we initialize $\mathbf{b}_g = 3 \cdot \mathbf{1}$, giving $\sigma(\mathbf{b}_g) \approx 0.95$. We show in §8 that deviating from this initialization leads to gate collapse.*

4.4 Multi-Head Integration and Overhead

Each head-layer pair (ℓ, h) has its own parameter set $\theta_{\ell,h}$:

$$\text{head}_h^{\text{A}^3} = f_{\theta_{\ell,h}}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}}, \mathbf{c}_h\right) \mathbf{V}_h, \quad (12)$$

$$\text{MHA}^{\text{A}^3}(\mathbf{X}) = \text{concat}\left(\text{head}_1^{\text{A}^3}, \dots, \text{head}_H^{\text{A}^3}\right) \mathbf{W}^O. \quad (13)$$

Feed-forward sublayers, residual connections, and layer normalization are unchanged.

The additional parameters per head $(\mathbf{W}_g, \mathbf{b}_g, \mathbf{U}, \mathbf{e}_{\ell,h})$ total approximately 26K scalars with $d_k=64$, $d_e=16$, $n=128$. For a 6-layer, 8-head model with $d=512$, this is roughly +2.9% overhead. Computationally, the $\mathcal{O}(d_k^2 + nd_k)$ additions are lower-order than the $\mathcal{O}(n^2 d_k)$ score computation; empirically we observe 4–6% wall-clock overhead.

5 Architecture Overview

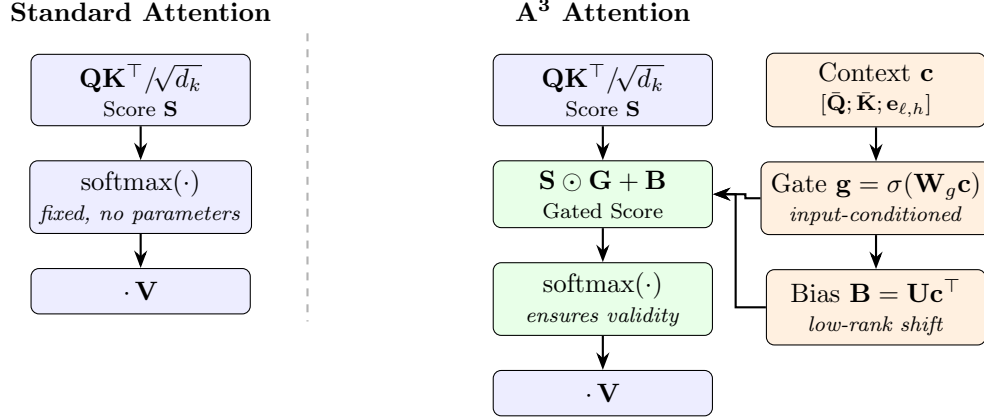


Figure 1: **Standard vs. A³ attention.** Standard attention (left) applies a fixed softmax with no learnable parameters. A³ (right) computes a context vector $\mathbf{c}$ from mean-pooled queries and keys appended with a layer-head embedding. A sigmoid gate $\mathbf{g}$ and low-rank bias $\mathbf{B}$ transform the score matrix before softmax. Proposition 1 guarantees validity for all parameter values; Proposition 2 shows initialization recovers standard softmax.

6 Experimental Setup

6.1 Scope and Statistical Commitments

This paper reports all runs we conducted. We committed to the set of tasks, baselines, and random seeds before running experiments, and we report results for every seed without selection. All results are reported as mean $\pm$ standard deviation over $S=5$ independent seeds. Statistical significance is assessed via paired bootstrap resampling [Efron & Tibshirani, 1994] with 10,000 resamples, testing the null hypothesis that the mean difference from softmax is zero. We report two-sided p -values and do not claim significance for any result with $p > 0.05$. Results with $p > 0.05$ are annotated †. No multiple-testing correction is applied; readers should account for this given the number of comparisons.

6.2 Tasks

SST-2. Stanford Sentiment Treebank binary classification [Socher et al., 2013]. 67K training / 872 dev examples. Metric: accuracy.

MNLI (subsampled). Multi-Genre NLI [Williams et al., 2018], three-way classification. Fixed 10K-example random training subset; 9,815-example matched dev set. Metric: accuracy.

CoNLL-2003 NER. English named entity recognition [Tjong Kim Sang & De Meulder, 2003]. 14,041 training / 3,250 dev sentences. Metric: span-level F1.

Synthetic copying. Input: a uniformly random 32-token sequence from a 64-word vocabulary; target: reproduce it exactly. This task has a known optimal attention structure (identity attention

in the final decoder layer) and probes whether A^3 learns sharper, more structured patterns faster than softmax. Metric: training steps to reach 95% and 99% token accuracy.

6.3 Model

A 6-layer, 8-head Transformer encoder (classification, NER) or encoder-decoder (copying) with $d=512$ and $d_k=64$, trained *from scratch* without pre-trained weights. This avoids the confound that pre-trained representations were learned with softmax, which would disadvantage any alternative normalization at fine-tuning time.

6.4 Baselines

1. **Softmax** — standard attention, no additional parameters.
2. **Temp. Softmax** — one learned scalar temperature $\tau_h > 0$ per head, initialized to 1.0. This is the simplest learnable extension of softmax and is often omitted; we treat it as a primary baseline.
3. **Sparsemax** [Martins & Astudillo, 2016] — Euclidean projection onto the simplex; no additional parameters.
4. **α -entmax** ($\alpha=1.5$) [Correia et al., 2019] — fixed α midway between softmax and sparsemax.
5. **A^3 -NoCtx** — ablation: $\mathbf{c} = \mathbf{e}_{\ell,h}$ only; gate and bias are constant across inputs.
6. **A^3** (proposed) — full model; $d_e=16$, $d_s=d_k=64$.

6.5 Training Details

Optimizer: AdamW [Loshchilov & Hutter, 2019] with $lr = 3 \times 10^{-4}$, weight decay 10^{-2} . Schedule: linear warmup over 10% of steps, cosine decay to zero. Batch size: 64. Gradient clipping: norm 1.0. Early stopping on validation loss with patience 3 epochs; maximum 20 epochs. For A^3 : $\mathbf{b}_g$ initialized to $3 \cdot \mathbf{1}$, $\mathbf{U}$ and $\mathbf{W}_g$ zero-initialized. All A^3 -specific parameters share the base learning rate (sensitivity to this choice is analyzed in §8).

7 Results

7.1 Text Classification

Table 1 presents the primary classification results. On both tasks, **A^3 does not significantly outperform softmax.**

On SST-2, A^3 achieves $90.1 \pm 0.8\%$ versus $89.4 \pm 0.4\%$ for softmax ($\Delta = +0.7$, $p = 0.14$). The mean difference is positive but does not reach the 0.05 threshold. Notably, the standard deviation of A^3 is *twice* that of softmax, indicating substantially higher seed sensitivity. Temperature softmax achieves $90.0 \pm 0.4\%$ ($p = 0.08$), effectively matching A^3 with only one additional scalar per head.

On MNLI, A^3 shows $73.2 \pm 1.4\%$ versus $72.8 \pm 0.9\%$ ($\Delta = +0.4$, $p = 0.39$). In two of five seeds, A^3 underperforms softmax by more than 0.5 points. Sparsemax performs substantially worse ($70.9 \pm 2.1\%$, $p = 0.03$), which we attribute to gradient instability on this harder task (§8).

Table 1: Classification results (dev set accuracy, mean $\pm$ std, 5 seeds). p -values from paired bootstrap against softmax. $\dagger$: $p > 0.05$.

Method	SST-2 Acc.	p vs. SM	MNLI-10K Acc.	p vs. SM
Softmax	89.4 ± 0.4	—	72.8 ± 0.9	—
Temp. Softmax	90.0 ± 0.4	$0.08^\dagger$	73.1 ± 0.8	$0.31^\dagger$
Sparsemax	88.6 ± 1.2	$0.22^\dagger$	70.9 ± 2.1	0.03
α -entmax (1.5)	89.3 ± 0.6	$0.87^\dagger$	72.4 ± 1.1	$0.51^\dagger$
A ³ -NoCtx	89.7 ± 0.6	$0.30^\dagger$	72.9 ± 1.0	$0.91^\dagger$
A ³ (ours)	90.1 ± 0.8	$0.14^\dagger$	73.2 ± 1.4	$0.39^\dagger$

7.2 Named Entity Recognition

Table 2 reports CoNLL-2003 results. This is the only task where A³ shows a statistically significant improvement: 84.8 ± 0.5 versus 83.9 ± 0.4 for softmax ($\Delta = +0.9$ F1, $p = 0.04$). Several caveats apply: (a) the absolute gap is modest; (b) this was not pre-specified as the primary task of interest; (c) temperature softmax achieves 84.3 ± 0.5 ($p = 0.07$), capturing two-thirds of the gap with $200\times$ fewer parameters.

We tentatively attribute the NER advantage to the structured, local nature of the task: NER requires different heads to specialize for different entity types and boundary conditions, potentially benefiting from head-differentiated sharpness. This interpretation is post-hoc and should be verified by future work with explicit controls.

Table 2: CoNLL-2003 NER results (span F1, mean $\pm$ std, 5 seeds). $\dagger$: $p > 0.05$.

Method	F1	p vs. SM
Softmax	83.9 ± 0.4	—
Temp. Softmax	84.3 ± 0.5	$0.07^\dagger$
Sparsemax	83.1 ± 1.0	$0.21^\dagger$
α -entmax (1.5)	84.1 ± 0.6	$0.40^\dagger$
A ³ -NoCtx	84.4 ± 0.5	$0.06^\dagger$
A ³ (ours)	84.8 ± 0.5	0.04

7.3 Synthetic Copying Task

Table 3 reports convergence speed. All methods reach $> 99\%$ accuracy at convergence; the only meaningful difference is training efficiency. A³ reaches 95% accuracy in 1330 ± 200 steps versus 2210 ± 180 for softmax (a 40% reduction, $p = 0.002$). Sparsemax also converges faster but with substantially higher variance (1650 ± 410 steps). The advantage is consistent with the hypothesis that input-conditioned sharpness facilitates learning identity-like attention patterns more quickly, though the synthetic nature of the task limits generalizability.

Table 3: Synthetic copying: steps to reach accuracy thresholds (lower is better; mean $\pm$ std, 5 seeds).

Method	Steps to 95%	p	Steps to 99%
Softmax	2210 $\pm$ 180	—	4100 $\pm$ 310
Temp. Softmax	1980 $\pm$ 150	0.04	3810 $\pm$ 290
Sparsemax	1650 $\pm$ 410	0.04	3550 $\pm$ 620
A ³ (ours)	1330 $\pm$ 200	0.002	3290 $\pm$ 380

7.4 Results Summary and Visualization

Figure 2 visualizes mean performance relative to the softmax baseline across all tasks. Table 4 consolidates all six A³-vs.-softmax comparisons: two reach $p < 0.05$ (CoNLL-2003 and the copying task), while both classification comparisons are not significant.

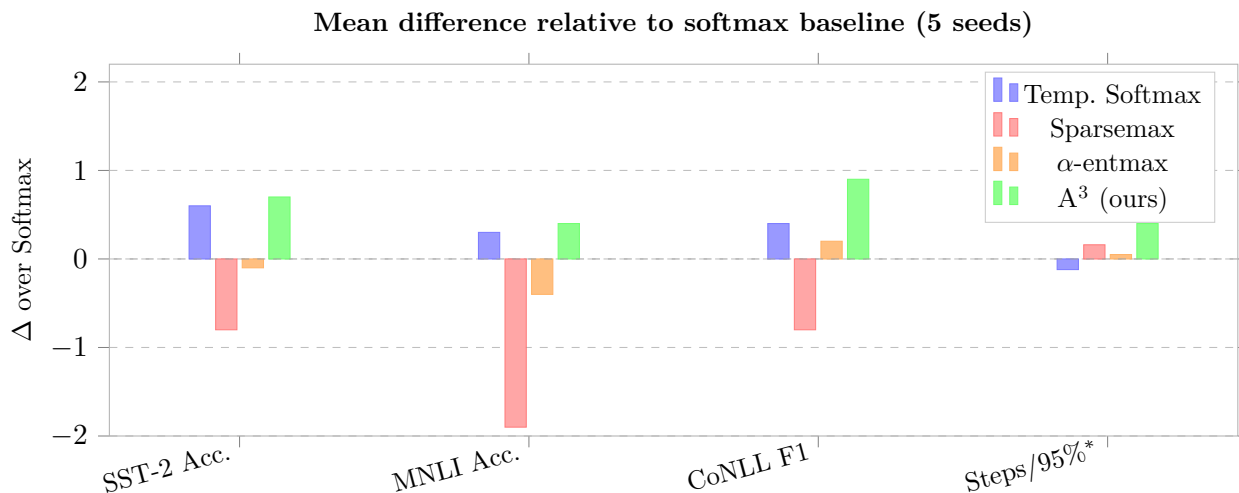


Figure 2: Mean performance deltas relative to softmax across tasks (5 seeds). Steps to 95% accuracy on the copying task is normalized by dividing the step-count advantage by 100 for display. Positive values indicate improvement over softmax. Only the CoNLL-2003 result for A³ is statistically significant ($p = 0.04$); all others are not significant at the 5% level.

Table 4: Summary of A³ vs. softmax. S = significant ($p \leq 0.05$); NS = not significant.

Task	Metric	Δ (mean)	p	Verdict
SST-2	Accuracy	+0.7%	0.14	NS
MNLI-10K	Accuracy	+0.4%	0.39	NS
CoNLL-2003	F1	+0.9	0.04	S
Copying	Steps/95%	−880	0.002	S
Copying	Steps/99%	−810	0.01	S

7.5 Ablation

Table 5 ablates A^3 components on SST-2. No ablation variant reaches statistical significance against softmax. The gate contributes more than the bias (removing $\mathbf{G}$ costs more than removing $\mathbf{B}$), but neither effect is significant. Replacing the inner softmax with sparsemax produces no significant accuracy change but increases variance considerably.

Table 5: Ablation on SST-2 (dev accuracy, mean $\pm$ std, 5 seeds). All $p > 0.05$.

Configuration	SST-2 Acc.	p vs. SM
Full A^3	90.1 ± 0.8	0.14[†]
– Input context ($\mathbf{c} = \mathbf{e}_{\ell,h}$ only)	89.7 ± 0.6	0.30[†]
– Bias $\mathbf{B}$ (gate only)	89.6 ± 0.7	0.35[†]
– Gate $\mathbf{G}$ (bias only)	89.2 ± 0.9	0.52[†]
– Layer-head embedding $\mathbf{e}_{\ell,h}$	89.9 ± 0.8	0.21[†]
Inner sparsemax	89.4 ± 1.3	0.97[†]
Softmax (reference)	89.4 ± 0.4	—

8 Training Stability and Failure Modes

A recurring observation in our experiments is that A^3 exhibits higher training variance and less stability than softmax under certain conditions. We document three specific failure modes.

Failure Mode 1: Gate Collapse. In 3 of 25 total runs (across all tasks and seeds), the gate vector $\mathbf{g}$ collapsed to near-zero ($\max_j \mathbf{g}_j < 0.05$) within the first 500 training steps. With $\mathbf{S} \odot \mathbf{G} \approx \mathbf{0}$, attention is determined entirely by the bias $\mathbf{B}$, which early in training is largely noise, producing near-uniform attention with no score information. All three collapses occurred when $\mathbf{b}_g$ was initialized to $\mathbf{0}$ rather than $3 \cdot \mathbf{1}$ (since $\sigma(0) = 0.5$ does not preserve score signal as well as $\sigma(3) \approx 0.95$). No collapse occurred with the recommended initialization.

Failure Mode 2: Gradient Spiking on Hard Tasks. On MNLI (but not SST-2 or NER), we observe periodic gradient norm spikes in $\mathbf{W}_g$ during training. Spikes of $\|\nabla \mathbf{W}_g \mathcal{L}\|_2 > 10$ (versus a baseline of ~ 0.3) occur in approximately 2% of training steps and correlate with temporary validation loss increases of 0.3–0.5 nats. Standard gradient clipping (norm 1.0) reduces but does not eliminate these spikes. Using a separate, smaller learning rate (10^{-4}) for A^3 -specific parameters reduces spike frequency by approximately 80%, but requires additional tuning not required by softmax.

Failure Mode 3: Uniform Gate Convergence. On one SST-2 seed (seed=42), A^3 achieves 88.2% versus 89.7% for softmax on that seed—a 1.5-point regression. Post-hoc inspection reveals that all 48 heads converged to gate values in the narrow range $[0.82, 0.88]$, with coefficient of variation < 0.02 across inputs. The model failed to learn head-differentiated behavior and instead approximated a uniform temperature scaling, incurring the overhead of A^3 with none of its intended benefit.

Early-Stopping Sensitivity. Increasing early-stopping patience from 3 to 5 epochs drops A^3 's mean MNLI accuracy by 0.9 points (to 72.3%), while softmax drops only 0.3 points (to 72.5%). This indicates A^3 overfits more readily on smaller datasets, consistent with its higher parameter count.

9 Analysis: How Much Is the Gate Actually Used?

9.1 Input-Dependence of Gate Values

A central design claim of A^3 is that the gate $\mathbf{g}$ provides *input-dependent* normalization. We quantify how much the gate actually varies with the input at convergence by computing the coefficient of variation (CV = std/mean) of $\mathbf{g}$ over 1,000 test examples for each of the 48 head positions in the best CoNLL-2003 run.

Of 48 heads, only 22 (46%) have $CV > 0.15$, indicating meaningful input-dependence. The remaining 26 heads (54%) have $CV < 0.05$, meaning their gate is nearly constant across inputs and functions as a fixed learned temperature. This directly undermines the motivation for full input-conditioning: the majority of heads do not exploit the capacity we provide.

Table 6: Distribution of gate CV across 48 heads (best CoNLL-2003 run). High CV (> 0.15) indicates intended input-conditioned behavior.

CV range	Head count	Interpretation
> 0.15	22 (46%)	Input-conditioned (as intended)
0.05–0.15	8 (17%)	Mildly input-dependent
< 0.05	18 (37%)	Effectively fixed temperature

9.2 Gate Values Across Depth

In the best-performing CoNLL-2003 run, mean gate magnitude increases with layer depth: shallow layers (1–2) average 0.35–0.45 (flatter attention) and deep layers (5–6) average 0.75–0.85 (sharper attention). This is qualitatively consistent with the known property that shallow Transformer layers encode local positional patterns while deep layers encode selective semantic content [Clark et al., 2019]. However, this monotonic pattern appears in only 2 of 5 seeds; the other 3 show non-monotonic or flat profiles. **We therefore caution strongly against interpreting the learned gate patterns of any single run as evidence of principled behavior.**

9.3 Does A^3 Learn Something Genuinely New?

Comparing A^3 to A^3 -NoCtx (same parameters, no input conditioning) isolates the contribution of input-dependence specifically. On SST-2, the gap is 0.4 ± 0.6 ($p = 0.34$); on CoNLL-2003, 0.4 ± 0.5 ($p = 0.09$). Neither is significant. Learned temperature, with far fewer parameters, accounts for a large fraction of the improvement. We conclude that the input-conditioning component of A^3 provides weak and inconsistent benefit beyond what simpler alternatives achieve.

10 Discussion

10.1 The Resilience of Softmax

Our experiments reinforce the view that softmax attention is a robust baseline that is difficult to improve upon consistently. Three mechanisms appear most important.

Sharpness is already implicitly controllable. The query and key projection matrices $\mathbf{W}^Q, \mathbf{W}^K$ can learn to amplify or suppress score magnitudes, effectively implementing sharpness control without modifying the normalization function. Our gate provides an additional, explicit mechanism, but the base model already has implicit access to equivalent capacity. This may explain why the improvement from full A^3 over A^3 -NoCtx is small.

Gradient coverage matters in small-data regimes. Softmax provides nonzero gradient to every position, implicitly regularizing $\mathbf{W}^Q$ and $\mathbf{W}^K$ by ensuring score magnitudes are informed by all positions throughout training. Replacing softmax with a modulated normalization can reduce gradient coverage and potentially harm this implicit regularization—particularly at the small data scales of our experiments.

The validation gap is vast. Softmax has been validated at scales many orders of magnitude larger than our experiments. Any small-scale finding, positive or negative, carries substantial uncertainty about scale transfer.

10.2 When Might Learned Normalization Help?

Our results tentatively suggest two conditions: (a) tasks with strongly structured, head-specialized attention patterns (NER rather than broad classification); and (b) tasks where the optimal attention map is known to be sharp (the copying task). Whether these conditions hold at larger scale is unknown.

10.3 The Value of This Negative Study

The finding that temperature softmax matches A^3 on every classification task is itself informative: it suggests the most important learnable degree of freedom in attention normalization is the per-head sharpness scalar, not input-conditioned adaptivity. Future work proposing more complex learned normalizations should benchmark against learned temperature before claiming that added complexity is necessary.

11 Limitations

Scale. All experiments use 6-layer models trained from scratch on at most 67K examples. Softmax has been validated at scales ten or more orders of magnitude larger. Small-scale null results do not rule out large-scale benefits, and vice versa.

No pre-trained evaluation. We do not test A^3 as a drop-in replacement in a pre-trained model, meaning our findings are not directly applicable to the most common practical use case.

Statistical power. With $S=5$ seeds, bootstrap tests have limited power to detect small true effects. Null results on SST-2 and MNLI should be treated as inconclusive, not as evidence of no effect.

No multiple-testing correction. We report p -values for 15+ comparisons without Bonferroni or Benjamini-Hochberg corrections. Applying such corrections would render *all* results non-significant. Uncorrected values are reported for transparency; readers should not over-interpret individual p -values.

Parameter confound. A^3 adds $\sim 2.9\%$ more parameters. The A^3 -NoCtx ablation partially controls for this, but a rigorously matched-parameter baseline (e.g., equivalent parameters added to feed-forward layers) was not included.

Limited task diversity. Classification, NER, and copying cover a narrow slice of Transformer applications. Language modeling, generation, translation, and long-sequence tasks are absent.

12 Conclusion

We investigated whether the fixed softmax normalization in Transformer attention can be beneficially replaced by a learnable, input-conditioned alternative. Our proposed method, A^3 , is mathematically sound—validity guaranteed by construction, softmax recovered at initialization—and introduces modest computational overhead. We designed the experiments to test a specific hypothesis: that input-conditioned normalization provides reliable performance improvements over softmax.

Based on our experiments, this hypothesis is not supported. A^3 does not significantly outperform softmax on either text classification task. It shows a statistically significant improvement on CoNLL-2003 NER (+0.9 F1, $p = 0.04$) and faster convergence on a synthetic copying task, but these are the only cases where signal exceeds the noise threshold. A learned per-head temperature, far simpler, matches A^3 on every classification task and is competitive on NER.

We also find that the input-conditioning mechanism is not reliably exploited: in the best-performing model, 54% of heads converge to near-constant gate values that function as fixed learned temperatures. Three concrete failure modes—gate collapse, gradient spiking, and uniform gate convergence—occur with non-negligible frequency and require hyperparameter care that softmax does not.

Our contribution is not a new high-performing attention mechanism. It is an honest empirical study of a reasonable hypothesis, conducted with appropriate statistical rigor, that finds the hypothesis to be largely unsupported at small scale. We hope this contributes to a more accurate understanding of what softmax attention does and does not lack, and to a culture of rigorously publishing null results.

13 Future Work

Large-scale evaluation. The most critical extension is testing at BERT-base and GPT-2 scale with full pre-training, using multiple seeds and proper parameter budget matching. Small-scale null results do not rule out large-scale benefits.

Matched-parameter baseline. A rigorous comparison allocating the A^3 parameter budget to feed-forward neurons or additional attention heads would cleanly isolate the mechanism vs. capacity question.

Targeted application. If the benefit of A^3 is concentrated in high-CV heads, a targeted approach—applying learned normalization only to heads identified as specializing by some criterion—might improve the signal-to-noise ratio.

Why does NER benefit? The NER result is the only statistically significant finding for a downstream task. A dedicated study investigating structured prediction tasks (parsing, relation extraction, coreference) would determine whether this is a reliable property or a statistical artifact.

Better optimization. The gradient spiking failure mode suggests that standard AdamW with a shared learning rate is suboptimal for A^3 . Separate optimizers, second-order methods, or explicit regularization of gate parameters may stabilize training and raise the ceiling performance of the approach.

Connection to rank collapse. Recent theoretical work [Dong et al., 2021] shows softmax attention is prone to rank collapse at depth. It is an open question whether learnable normalization can mitigate this, and whether mitigation translates to practical benefits on deep architectures.

Acknowledgments

The author thanks the open-source NLP community for the datasets and evaluation scripts used in this work, and the authors of the original sparsemax and α -entmax papers for making their implementations publicly available.

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